🔗 Topic 1 at a glance: which bones can we trust?

- **Superimposition** lines up two X-rays on structures that do not change; whatever still differs is growth or treatment.
- **Gold standard:** Björk's metal implants in 2D; bone plates and screws in 3D today.
- **Maxilla:** register on the **zygomatic process (key ridge)**. The palate sinks and its plane tilts, so palatal "best fit" at ANS hides **30–50 %** of eruption.
- **Mandible:** register on the **chin and inner symphysis**; the third-molar crypt and the mandibular canal move with growth.

An Investigation of Maxillary Superimposition Techniques Using Metallic Implants

by **Doppel, Damon, Joondeph and Little** · Am J Orthod Dentofacial Orthop 1994;105(2):161–168 · University of Washington

🎯 Study purpose

To find **stable anatomic landmarks in the maxilla** that can be used reliably for cephalometric superimposition, judged against **metallic implants (gold standard)**.

🔬 Study design

- **50 patients** (23 M, 27 F), chosen from about **700** implant records; most were treated (edgewise, headgear, elastics; extraction or not).
- Age at T1 **11.9 ± 1.4 y**, at T2 **16.0 ± 1.7 y**; interval **3.0–6.4 y** (mean **4.2 ± 0.96 y**).
- **Implant sites (Björk):** below ANS (anterior), zygomatic process (lateral), palate–alveolar border medial to the first molar.
- **Selection:** all three implants in perfect orientation, films ≥3 years apart, high quality, same cephalostat and head orientation.
- One operator; tracing error **0.5 mm**. Films superimposed on the implants; the maximum drift of **17 structures** measured with calipers.

📊 Key results: drift from implant superimposition (Table II)

Structure	Mean ± SD, mm	Note
Internal palatal structure	0.85 ± 0.43	Best number, poor fit by eye: limited value
Anterior zygomatic process	**0.95 ± 0.57**	Reliable; traceable in 80 %; direction varies (27.5 % forward, 35 % back, 37.5 % none)
Posterior zygomatic process	**0.97 ± 0.43**	Reliable; traceable in 100 %; always moves back (apposition)
Palate, superior / inferior border	1.05 ± 0.52 / 1.14 ± 0.60	Not reliable
ANS, vertical / horizontal	1.29 ± 0.86 / 1.43 ± 1.14	Not reliable
Orbital floor	1.40 ± 0.98	Rises with growth
Infraorbital foramen	1.67 ± 1.31	Not reliable; text says traced in 32 %, Table II has 34 of 50
A point	2.13 ± 1.63	Not reliable; moves with tooth movement
PNS	2.27 ± 1.75	Not reliable; visible in about half; lengthens
PTM, vertical / horizontal	2.39 ± 1.69 / 1.52 ± 0.99	Very inaccurate

🔄 Rotational changes against the implant line (Table III)

Plane	Mean	SD	Range
Palatal plane (ANS–PNS)	0.92°	2.10°	−5° to +8°
S-N line	−0.56°	2.19°	−6° to +9°

The maxilla rotates against the cranial base, and the palatal plane rotates inside the maxilla, differently in each patient.

🧠 Main conclusions

1. The **anterior and posterior zygomatic contours** fit the implants best, about 1 mm.
2. **Vertical remodelling:** the orbital floor rises more than the palatal plane lowers, 1.5 : 1.
3. The maxilla and the palatal plane rotate variably.
4. Internal palate of limited value; **infraorbital foramen, PTM, PNS, ANS, A point and the palate borders are not reliable**.
5. No maxillary area is completely reliable.

🦷 Clinical implications

- **Palatal plane at ANS:** implants appear to move up and back, incisors look retracted, molars descend, the occlusal plane rotates up and forward.
- **PTM registration:** teeth look labially moved with little descent.
- **Best approximation:** register on the **anterior and posterior zygomatic contours**, letting the orbital floor rise 1.5 × the palate's lowering.

✅ Number	Meaning
50 of ~700	patients usable
4.2 ± 0.96 y	mean interval between films
0.95 / 0.97 mm	anterior / posterior zygomatic drift
−5° to +8°	palatal-plane rotation range
1.5 : 1	orbital-floor rise : palatal-plane lowering

Maxillary Superimposition: A Comparison of Three Methods for Cephalometric Evaluation of Growth and Treatment Change

by **Ib Leth Nielsen** · Am J Orthod Dentofacial Orthop 1989;95(5):422–431 · University of California, San Francisco

📚 Study purpose

To compare three maxillary superimposition methods (**best fit, implant and structural**) and to test the validity and reliability of the **structural method**.

🔬 Sample and methods

- **18 subjects** (6 boys, 12 girls), usable from **35** implant subjects; selected on film quality.
- Age at T1 **10 ± 0.37 y**; interval **4 ± 0.56 y**; treated and untreated mixed; mostly Class II, no Class III.
- At least **3–4 stable implants** in the zygomatic process on both films.
- Separate tracings for each method, implants hidden for the anatomic methods; double determinations; a second observer repeated the structural method.
- Reference axes: **X = Downs' occlusal plane**, **Y = perpendicular through the first-molar mesial cusps**.

🔀 The three methods

1. **Best fit:** palatal outlines aligned, registered at **ANS**. The traditional method; ignores nasal-floor resorption.
2. **Implant:** registered on the implants. The **gold standard**.
3. **Structural (Björk–Skieller):** registered on the **anterior surface of the zygomatic process**, with orbital-floor apposition set **equal** to nasal-floor resorption.

📊 Key results: vertical displacement over 4 years (Table II, mm)

Method	Molar cusp U6C	Molar apex U6A	Incisor edge UIE	Incisor apex UIA
Best fit	−3.40	−2.33	−1.40	−2.60
Implant	**−4.75**	**−3.80**	**−3.39**	**−4.36**
Structural	−4.99	−4.02	−3.39	−4.48

- **Best fit vs implant:** significantly different at every landmark (all P < .01, most P < .001): best fit under-reads eruption about 30 % at the molars and 50 % at the incisors.
- **Structural vs implant:** **no significant vertical difference** (slight overestimate); horizontally a systematic **≈ 0.5 mm shift** against the implants.
- **Implant skeletal changes (Table I):** ANS −1.62, PNS −0.70, A point −2.33 mm: **ANS lowered about twice as much as PNS**, because nasal-floor resorption is greater anteriorly. PNS moved back **2.3 mm** (best fit showed 1.2 mm).
- **Second observer (structural):** differences only vertically, at ANS, A point and the incisor points.

🧠 Conclusions and clinical implications

- The **structural method is valid and reliable**, especially vertically, but needs good films with minimal double contours (only 18 of 35 were usable).
- **Best fit** is not reliable for judging treatment outcome: it hides eruption and vertical growth.
- **PTM registration (Moore)** and the **infratemporal-fossa method (Riedel)** add error, because the maxilla lengthens at PNS and the posterior zygomatic surface is appositional; **Luder's key-ridge registration** also under-reads eruption.
- The implant method is assumed to be the most accurate; the structural method is the alternative for patients without implants.

✅ Number	Meaning
18 of 35	subjects usable
≈ 30 % / ≈ 50 %	eruption missed by best fit: molars / incisors
−4.75 vs −3.40 mm	molar-cusp descent: implant vs best fit
1.62 vs 0.70 mm	ANS vs PNS lowering with implants
≈ 0.5 mm	structural method's horizontal shift

Three-Dimensional Mandibular Regional Superimposition in Growing Patients

by **Nguyen, Cevidanes, Franchi, Ruellas and Jackson** · Am J Orthod Dentofacial Orthop 2018;153(5):747–754 · UNC Chapel Hill

📚 Study aims

1. Identify **stable mandibular structures in 3D** in growing patients, using **bone plates and screws** as reference, like Björk's implants.
2. Test the **reproducibility** of 3D voxel-based mandibular regional superimposition on those structures.

🔬 Materials and methods

- **Aim 1 (stability):** 20 Class III bone-anchor patients, **12.1 ± 1.3 → 13.3 ± 1.1 y**; plates and screws near the incisor apices; T1 and T2 registered on the plates.
- **Aim 2 (reproducibility):** 25 growing patients, **12.7 ± 1.4 y**, CVM 2–5 (10 Herbst, 10 Class II elastics, 5 Class III bone anchors); scans **12.6 ± 0.9 months** apart; two observers registered on chin + symphysis.
- CBCT (i-CAT or NewTom 3G, **0.3 mm voxels**); voxel-based registration in Slicer CMF; 3D surface models; absolute mean surface error.
- Excluded: craniofacial syndromes, marked asymmetry, motion artefacts. The **mandibular canal** was excluded in advance because it changes with growth.

📍 Regions evaluated

1. **Chin:** pogonion to B point, laterally to the lateral incisors.
2. **Symphysis:** inner cortex, from the inferior border up to the mental foramina.
3. **Third-molar crypts:** inferior contour of the third-molar germs, part of Björk's 2D list of stable regions.

📊 Key results: registration error against the bone plates (Table)

Region	Mean error, mm	SD	P
Chin	**0.37**	0.16	.765
Symphysis	**0.40**	0.14	.698
Third-molar crypt	**1.94**	0.06	.0025

Chin and symphysis are stable. **The third-molar crypt is unstable**, especially with vertical growth.

- **Reproducibility (chin + symphysis):** two observers differed by **0.12 ± 0.11 mm** on average, below the 0.3 mm voxel; ICC 0.998 (95 % CI 0.995–1.000).
- **Remodelling seen:** chin contour stable; deposition at the inferior symphysis (about 2 mm near menton); resorption of the anterior ramus and deposition posteriorly, as Björk and Enlow described.

🧠 Conclusions and clinical implications

1. **Chin and symphysis** are stable regions for 3D mandibular superimposition in growing patients.
2. Registering on them combined is **reliable and reproducible**.
3. 3D removes magnification error, lets head-position error be corrected easily, and uses thousands of voxels instead of a few landmarks.
4. Unlike Björk's 2D list, the **third-molar crypt** showed a large error (1.94 mm); the **canal** was excluded in advance because Björk himself found that it changes with growth.
5. Limits (our reading, not listed by the authors): one year of growth and small samples; the authors raise the concern that the plates sit close to the chin and argue against it (menton still remodelled about 2 mm).

✅ Number	Meaning
0.37 / 0.40 mm	chin / symphysis error
1.94 mm	third-molar crypt error
0.12 mm, ICC 0.998	agreement between two observers
0.3 mm	voxel size